

Pneumonia Classification Challenge

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Repository: https://github.com/yomnaeltnikety/ENG2440_A1

Task A: Data inspection

Input, output and clinical question

The input is a frontal chest radiograph in DICOM format, drawn from the RSNA Pneumonia Detection Challenge dataset [1]. The output is a binary prediction: Target = 1 means expert-annotated lung opacity consistent with possible pneumonia, and Target = 0 means no such opacity. Lung opacity is used because pneumonia can fill normally air-containing lung spaces with inflammatory material, producing radiographic opacity. This is an imaging proxy, not a confirmed clinical diagnosis [2].

The clinical question is: given a frontal chest X-ray, can a model classify whether the image contains lung opacity consistent with possible pneumonia?

Class Counts and metadata

There were 25,684 labelled examinations and 11,171 source patients from the underlying NIH ChestX-ray8 cohort [3]. The cohort was imbalanced: 20,025 examinations were negative and 5,659 were positive.

Table 1: Dataset summary and available metadata.

Quantity	Count	Proportion
Labelled examinations	25,684	100.00%
Source patients	11,171	—
Target = 0	20,025	77.97%
Target = 1	5,659	22.03%
AP / PA projection	11,705 / 13,979	45.57% / 54.43%

All inspected images were 1024×1024 pixels with MONOCHROME2 photometric interpretation. Projection was available as anterior-posterior (AP) or posterior-anterior (PA).

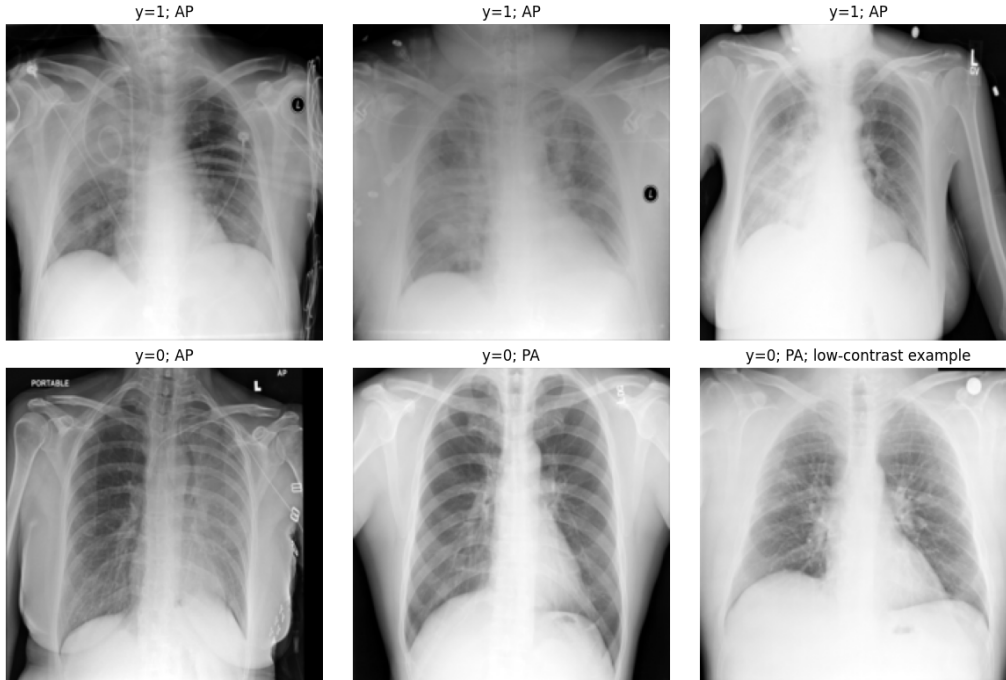


Figure 1: Representative positive and negative radiographs, including an unusual low-intensity-variation example. Labels are opacity annotations, not confirmed diagnoses.

Task B: Leakage-Safe Splits

The RSNA `patientId` is an examination identifier, so I joined it to the supplied NIH source-patient identifier and split by `nih_patient_id`. Assert statements confirmed that train, validation and test patient sets were pairwise disjoint. The test set was used only after model and threshold choices had been made.

Table 2: Patient-level split summary.

Split	Examinations	Patients	Positives	Prevalence
Train	18,257	7,819	3,996	21.89%
Validation	3,739	1,676	820	21.93%
Test	3,688	1,676	843	22.86%

One possible remaining leakage route is site leakage where a model could learn acquisition style if the hospital metadata are correlated with labels. I would test this using metadata balance, held-out-site evaluation and duplicate-image checks.

Task C: Preprocess and Augment

DICOM-to-tensor pipeline

DICOM pixels were converted to float, rescaled using the intercept, inverted for `MONOCHROME1` if needed, clipped between the 0.5th and 99.5th percentiles, rescaled to $[0, 1]$, quantised to 8-bit grayscale, resized with aspect ratio preserved and centred on a 224×224 canvas. The grayscale image was repeated into three channels and normalised with ImageNet mean and SD for ResNet-18.

Training augmentation

Training images received small rotations up to 5° , translations up to 3%, and contrast adjustments of 10%. These model positioning and exposure variation without changing anatomy. I did not use horizontal flips, large crops or anatomical deformations.

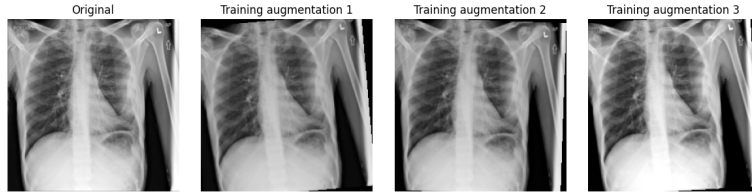


Figure 2: Training-only augmentation examples. Validation/test inputs use deterministic preprocessing.

Class imbalance

The training positive fraction was 21.89%. Weighted BCE used positive weight $(18257 - 3996)/3996 \approx 3.57$. On validation, unweighted BCE had slightly higher AP and better Brier score, while weighted BCE increased sensitivity but reduced specificity and precision.

Table 3: Class-imbalance comparison on validation for seed 2440 at threshold 0.5.

Loss	AP	Brier	Sensitivity	Specificity	Precision
Unweighted BCE	0.688	0.110	0.574	0.919	0.666
Weighted BCE	0.687	0.153	0.821	0.766	0.497

Because AP was the primary validation metric, the unweighted configuration was used for the headline CNN.

Task D: Simple Baseline

Majority baseline

The majority class in training was `Target = 0`, so the trivial baseline always predicted negative. On the test set it achieved 77.14% accuracy, 0 sensitivity and 1.00 specificity.

Linear feature baseline

The simple PyTorch baseline was `nn.Linear(6, 1)` using image mean, standard deviation, and the 10th, 25th, 75th and 90th intensity percentiles. Feature standardisation was fitted on training data only. Four validation trials were run; the best setting was learning rate 0.03 and weight decay 0, with validation AP 0.436.

Task E: Pretrained CNN

An ImageNet pretrained ResNet-18 [4] was adapted by replacing the 1000-class head with one output logit. The one-channel chest image was repeated into three channels. The model was trained first as a fixed feature extractor, then `layer4` and the head were fine-tuned. The same split and evaluation procedure were used for all models. Seeds 2440, 2441 and 2442 were used, Python, NumPy and PyTorch generators were seeded.

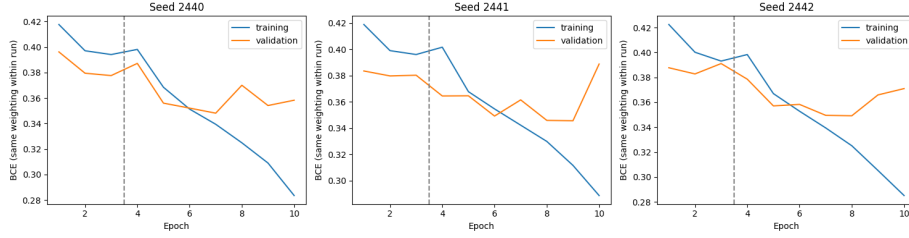


Figure 3: CNN training and validation losses. Seeds 2441 and 2442 show late validation loss increase while training loss falls, consistent with some overfitting.

Across the three CNN seeds, test AP was mean 0.7035 with sample SD 0.0068. The improvement over the linear baseline AP was 0.3023, larger than the observed one-SD seed variation.

Task F: Evaluate Performance

At threshold 0.5, the CNN improved sensitivity, F1, ROC-AUC and AP compared with both baselines. Precision is undefined for the majority classifier because it predicted no positives.

Table 4: Test-set performance at threshold 0.5.

Model	TN	FP	FN	TP	Acc	Sens	Spec	Prec	F1	ROC-AUC	AP
Majority	2845	0	843	0	0.771	0.000	1.000	—	0.000	0.500	0.229
Linear	2835	10	833	10	0.771	0.012	0.996	0.500	0.023	0.698	0.401
CNN 2440	2616	229	342	501	0.845	0.594	0.920	0.686	0.637	0.879	0.706
CNN 2441	2657	188	394	449	0.842	0.533	0.934	0.705	0.607	0.876	0.696
CNN 2442	2591	254	338	505	0.839	0.599	0.911	0.665	0.630	0.879	0.709

ROC-AUC measures ranking of positives above negatives. Precision-recall curves are more informative for this imbalanced cohort because they focus on positive retrieval, AP is the primary metric.

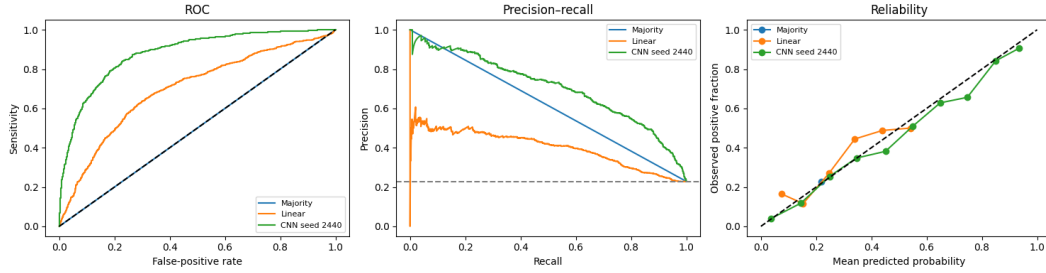


Figure 4: ROC, precision-recall and reliability curves for majority, linear and CNN seed 2440.

Two thresholds were selected using validation data only. The screening threshold maximised specificity while requiring validation sensitivity ≥ 0.90 , the high-specificity threshold maximised sensitivity while requiring validation specificity ≥ 0.90 .

Table 5: Frozen-threshold test performance.

Model	Rule	Threshold	Sensitivity	Specificity	F1
Linear	Screening	0.141	0.896	0.250	0.405
Linear	High-specificity	0.313	0.302	0.896	0.366
CNN 2440	Screening	0.131	0.900	0.659	0.590
CNN 2440	High-specificity	0.449	0.632	0.904	0.646

Calibration was assessed with a reliability curve and Brier score. Discrimination describes ranking ability, while calibration describes whether predicted probabilities match observed frequencies.

Task G: Interpretation and Error Analysis

For CNN seed 2440, representative TP, TN, FP and FN examples were selected near the 0.5 threshold where available, both correct and incorrect predictions were included. Grad-CAM used `layer4[-1]`, the final ResNet-18 residual block, because it contains high-level spatial features before pooling and classification.

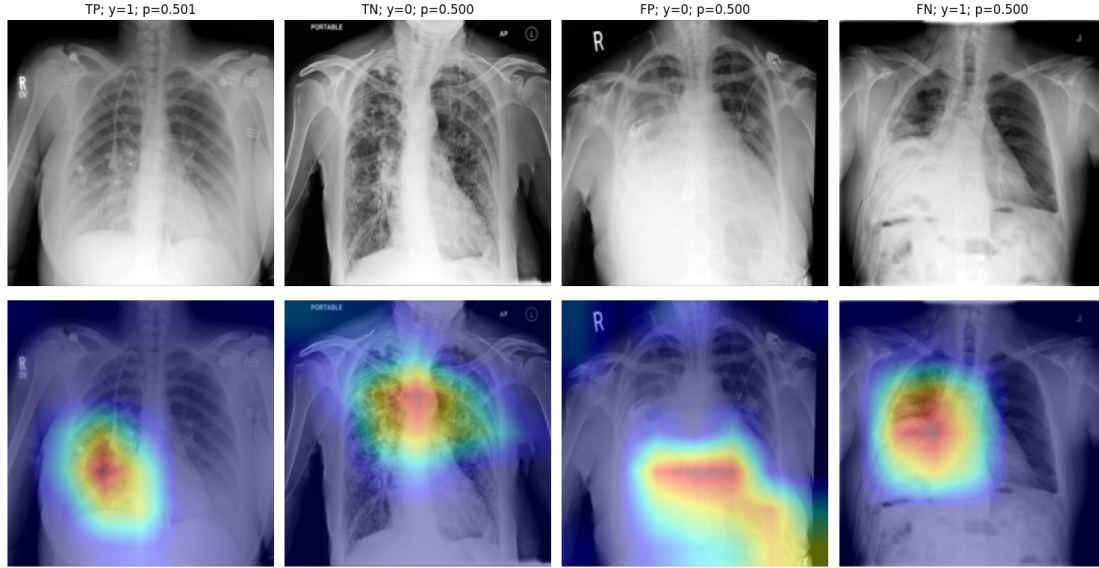


Figure 5: Selected predictions and Grad-CAM overlays. Grad-CAM explains positive-logit evidence and does not prove causation.

Table 6: Subgroup performance for CNN seed 2440 at threshold 0.5.

Field	Group	n	Prev	Sens	Spec	Prec	AP
View	AP	1701	0.389	0.698	0.826	0.719	0.775
View	PA	1987	0.091	0.215	0.973	0.448	0.348
Sex	F	1527	0.207	0.570	0.923	0.659	0.658
Sex	M	2161	0.244	0.609	0.917	0.702	0.734

AP examinations had much higher prevalence than PA examinations, so subgroup differences are descriptive and may reflect case mix or acquisition effects. Border CAM fractions for four displayed examples were 0.073, 0.089, 0.243 and 0.018, below the border area fraction 0.296, suggesting these examples were not dominated by border saliency. This is descriptive only and does not prove clinically meaningful localisation.

Bonus 1: Prevalence Shift and Threshold Transfer

Method. Two additional evaluation cohorts were constructed from the existing test split by randomly subsampling one class, without replacement, to obtain approximately 10% and 40% positive prevalence. The trained CNN and its validation selected screening threshold were kept fixed: neither retraining or threshold adjustment was performed.

Table 7: Performance under prevalence shift using the same trained CNN and frozen screening threshold.

Cohort	Prevalence	Sensitivity	Specificity	PPV	F1	ROC-AUC
Original	22.86%	0.900	0.659	0.439	0.590	0.879
Lower	10.00%	0.908	0.659	0.228	0.365	0.884
Higher	40.00%	0.900	0.659	0.637	0.746	0.876

Interpretation. Table 7 shows that sensitivity, specificity and ROC-AUC remained approximately stable across the original, 10% and 40% prevalence cohorts where sensitivity was 0.900, 0.908 and 0.900, specificity was 0.659 throughout and ROC-AUC was 0.879, 0.884 and 0.876. In contrast, precision fell from 0.439 to 0.228 at lower prevalence and rose to 0.637 at higher prevalence and F1 similarly changed from 0.590 to 0.365 and 0.746.

Sensitivity = $TP/(TP + FN)$ and specificity = $TN/(TN + FP)$ condition on positive and negative cases, respectively. Random subsampling preserves these rates in expectation, although finite sample variation can affect measured values. ROC-AUC measures positive-negative ranking and is therefore invariant to prevalence when the class conditional score distributions remain unchanged. However, precision = $TP/(TP + FP)$ combines both classes so that lower prevalence increases false positives relative to true positives, reducing the reliability of positive alerts. Since $F_1 = 2PR/(P + R)$, where P is precision and R is recall, declining precision lowers F1 despite stable recall.

If an emergency-department population had 10% prevalence rather than approximately 23%, these operating characteristics would imply about 3.4 false alerts per true-positive alert, compared with 1.3 originally. Positive alerts would therefore be substantially less trustworthy despite similar ranking performance. Resampling the same population isolates prevalence effects and it does not capture changes in patients, scanners or disease presentation that an actual deployment could introduce.

Conclusion

The CNN outperformed the feature baseline on the fixed patient-level test set, with mean AP 0.7035 across three seeds versus 0.401 for the linear baseline. The main limitations are proxy labels, possible acquisition shortcuts, resizing from 1024^2 to 224^2 , and no external validation.

AI Use Declaration

See the submitted `ai_use_declaration.txt` in the repository link. AI assistance was used for explanation, debugging support and report-formatting help and all code, outputs and interpretations were checked against the executed notebook.

References

1. Radiological Society of North America. RSNA Pneumonia Detection Challenge, 2018.
2. Shih G. et al. Augmenting the National Institutes of Health Chest Radiograph Dataset with Expert Annotations of Possible Pneumonia. *Radiology: Artificial Intelligence*, 2019.
3. Wang X. et al. ChestX-ray8: Hospital-scale Chest X-ray Database and Benchmarks. CVPR, 2017.
4. He K. et al. Deep Residual Learning for Image Recognition. CVPR, 2016.